

Matter Wave Manipulation

Topics

- Matter Wave Tractor Beam
- Analogy with Optical Manipulation
- Implementation on COMSOL Multiphysics
- An Optical Manipulation Technique
- Lateral Force by Matter Wave
- Results: Three Different Scenarios
- Application

Introduction: The Matter Wave Tractor Beam

Bessel beam (a structured beam) pulls the target particle.

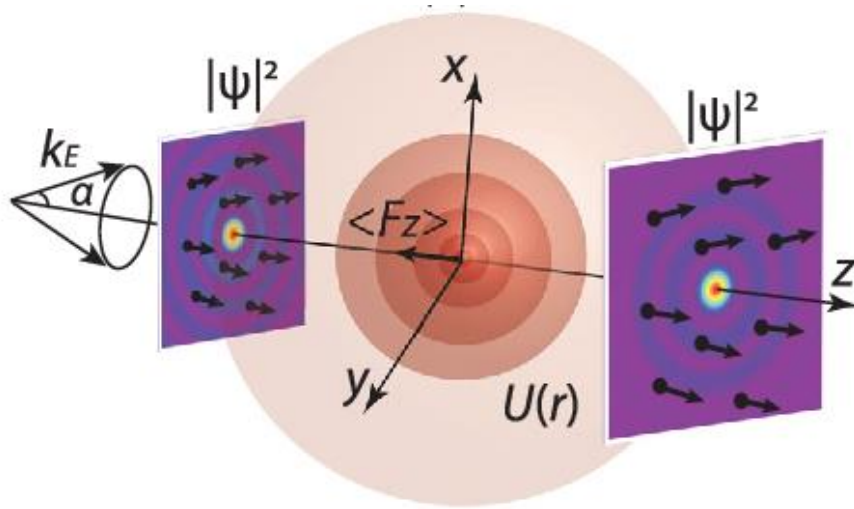


Figure: Reprinted from Gorlach, Alexey A., et al. "Matter-wave tractor beams." *Physical Review Letters* 118.18 (2017): 180401.

Matter wave momentum in forward axial direction:
Incoming waves < Outgoing waves

Momentum conservation: target particle gains momentum backward



Source: 

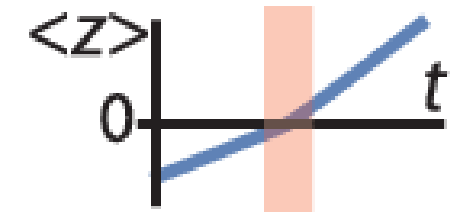


Figure: Reprinted from Gorlach, Alexey A., et al. "Matter-wave tractor beams." *Physical Review Letters* 118.18 (2017): 180401.

Analogy with Optical Manipulation

$$\langle \mathbf{F}_{\text{Total}}^{\text{Out}} \rangle = \oint \langle \bar{\mathbf{T}}^{\text{out}} \rangle \cdot d\mathbf{s}$$

Optical Stress Tensor (Ref. 3)

$$\langle \bar{\mathbf{T}}^{\text{out}} \rangle = \frac{1}{2} \text{Re} \left[\mathbf{D}_{\text{out}} \mathbf{E}_{\text{out}}^* + \mathbf{B}_{\text{out}} \mathbf{H}_{\text{out}}^* - \frac{1}{2} \bar{\mathbf{I}} (\mathbf{E}_{\text{out}}^* \cdot \mathbf{D}_{\text{out}} + \mathbf{H}_{\text{out}}^* \cdot \mathbf{B}_{\text{out}}) \right].$$

where,

Out represents total (incoming + outgoing) exterior fields

E : Electric field

D : Displacement vector,

H : magnetic field

B : induction vector

$$V = \frac{w}{k} = \frac{1}{\sqrt{\mu \epsilon}} \Rightarrow k = \frac{\sqrt{\mu \epsilon}}{w}$$

QM Stress Tensor (Ref. 1)

$$\bar{T} = \frac{\hbar^2}{2m} (\vec{\nabla} \psi \otimes \vec{\nabla} \psi^* + \vec{\nabla} \psi^* \otimes \vec{\nabla} \psi) + \bar{I} \left(((E - U_{\text{surf.}}) - U_{\text{sc.}}) |\psi|^2 - \frac{\hbar^2}{2m} |\vec{\nabla} \psi|^2 \right)$$

where,

ψ : stationary wave function,

$U_{\text{sc.}}$: potential energy of the incident particle due to the target

E and m : energy and mass of the incident particle.

$U_{\text{surf.}}$ is only applicable for our model, and will be defined in a later slide.

$$k = \frac{\sqrt{2m(E - V)}}{\hbar}$$

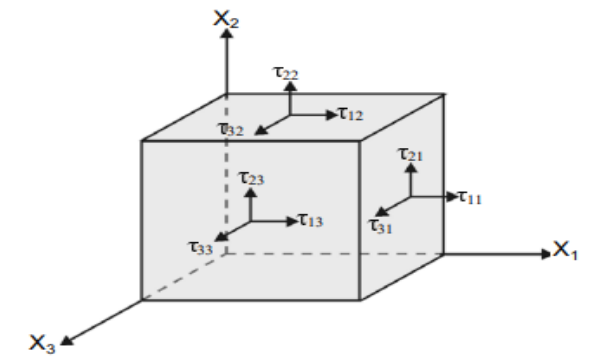


Figure: Reprinted from 2.20 - Marine Hydrodynamics, Spring 2005 Lecture 3

$$E_x \rightarrow \psi, \quad \mu \rightarrow m, \quad \epsilon \rightarrow 2(E - V), \quad \omega \rightarrow 1/\hbar. \quad (\text{Ref. 4})$$

Implementation on COMSOL Multiphysics

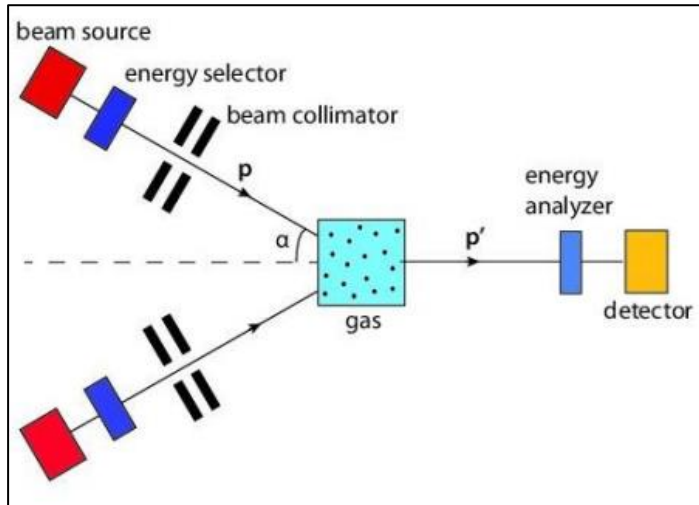
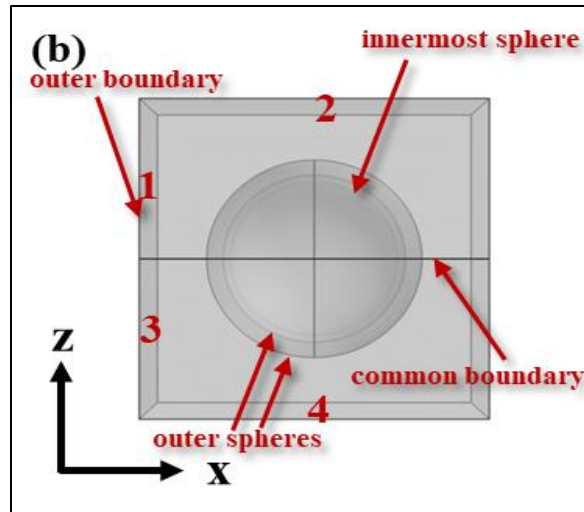
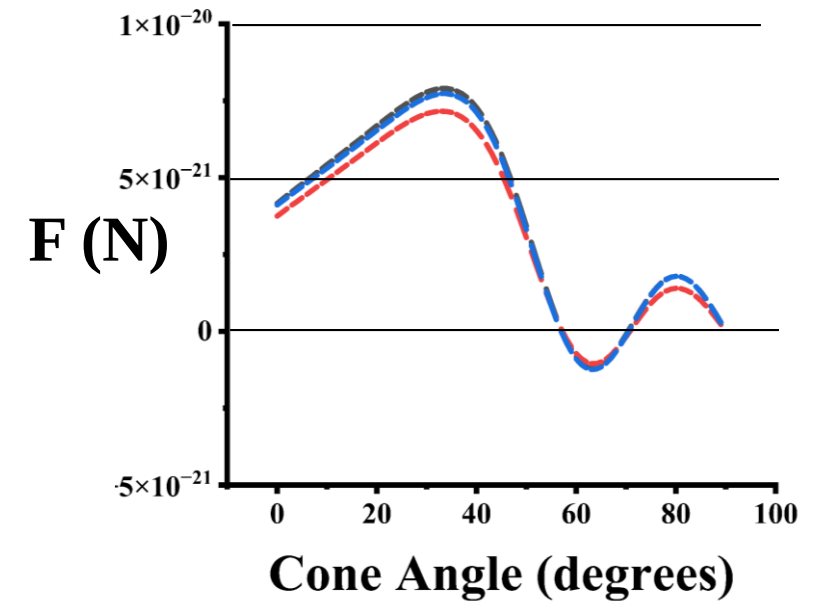


Figure: Reprinted from Gorlach, Alexey A., et al. "Matter-wave tractor beams." *Physical Review Letters* 118.18 (2017): 180401.



COMSOL Set-up



Optical Manipulation Technique

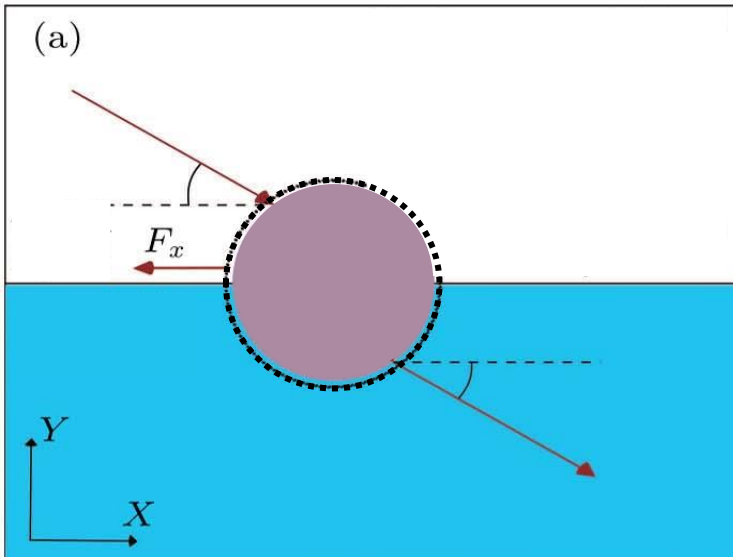


Figure: Adapted from Mahdy, M. R. C., et al. "Dielectric or plasmonic Mie object at air-liquid interface: The transferred and the traveling momenta of photon." *Chinese Physics B* 29.1 (2020): 014211.

light wave momentum in the forward axial direction:
Incoming waves < Outgoing waves

Momentum conversation: target particle gains momentum backward

This manipulation was resultant from inhomogeneous background
(refractive index)

Quantum Mechanical Lateral Force on an Atom due to Matter Wave

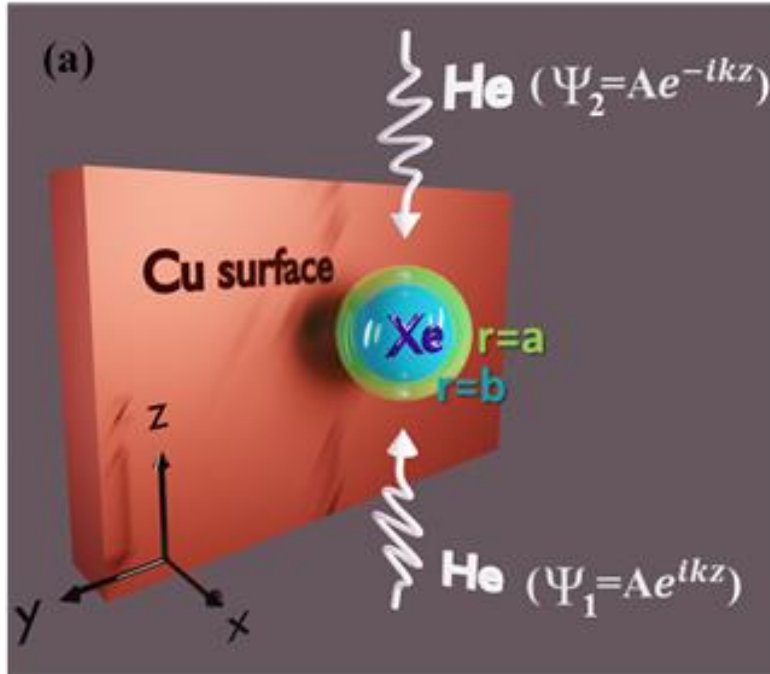


Figure:

- Two counter-propagating matter waves (Helium)
- target atom (Xenon)
- metal surface (Copper)

In optical manipulation, asymmetric properties of incident light (e.g. spin) and/or material of the target (e.g. chirality) result in lateral force.

We used the inhomogeneity of the background potential energy due to the surface.

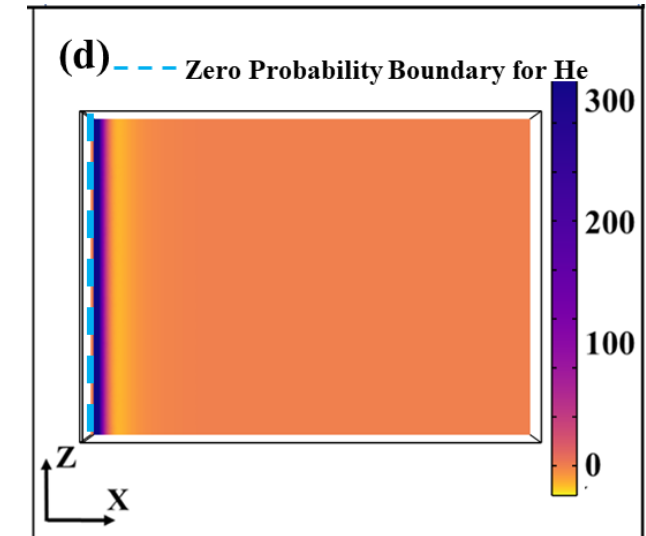
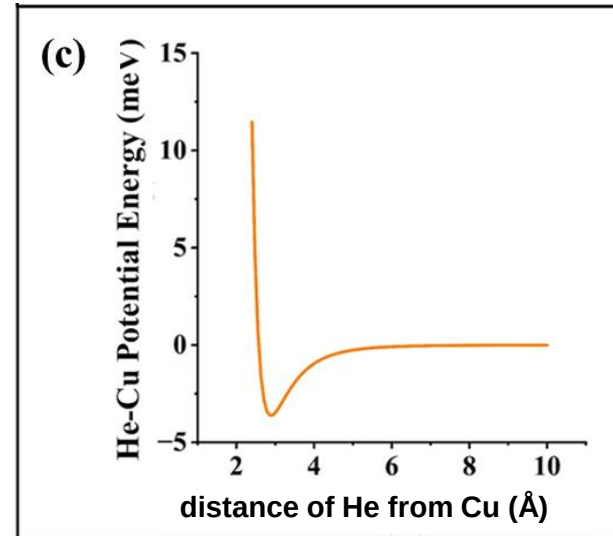
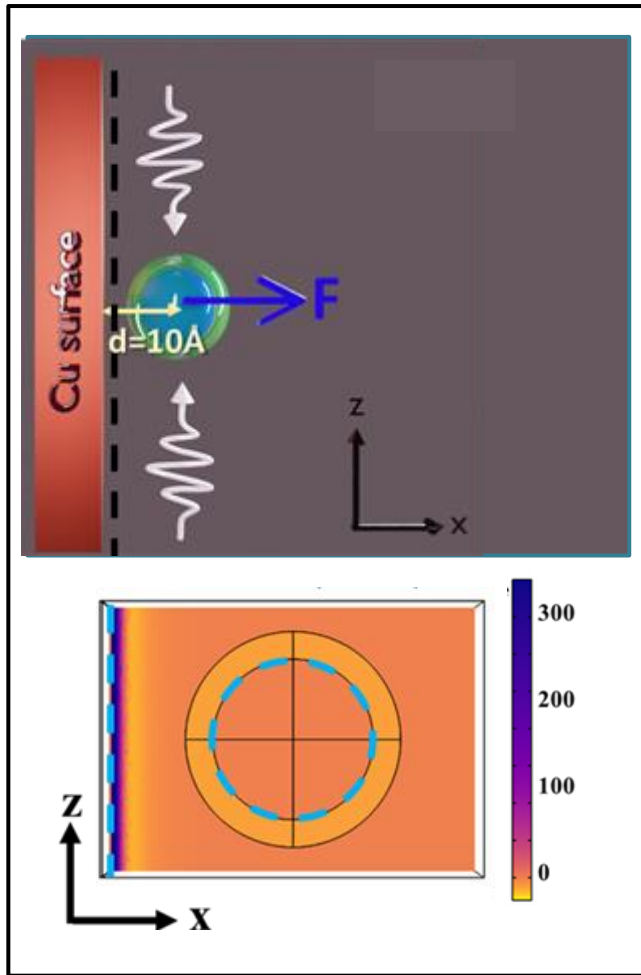


Figure: He-Cu Potential energy in COMSOL

Result: Three Different Scenarios

Depending on how close the target atom is to the surface, the force direction changes.



$d = 10 \text{ \AA}$

Total wave:

$$\psi(r, \theta) = A \left[e^{ikz} + e^{-ikz} + \underbrace{\left[\sum_{l=0}^{\infty} (2l+1) \frac{1}{k} e^{2i\delta_l} \sin(\delta_l) P_l(\cos \theta) + \sum_{l=0}^{\infty} (2l+1) \frac{1}{k} e^{2i\delta_l} \sin(\delta_l) P_l(\cos \theta) (-1)^l \right]}_{\text{Scattering amplitude}} \right] \frac{e^{ikr}}{r}$$

Scattering amplitude: $f_{\text{overall}}(\theta) = \sum_{l=0}^{\infty} (2l+1) \frac{1}{k} e^{2i\delta_l} \sin(\delta_l) P_l(\cos(\theta)) [1 + (-1)^l]$

Scattered wave:

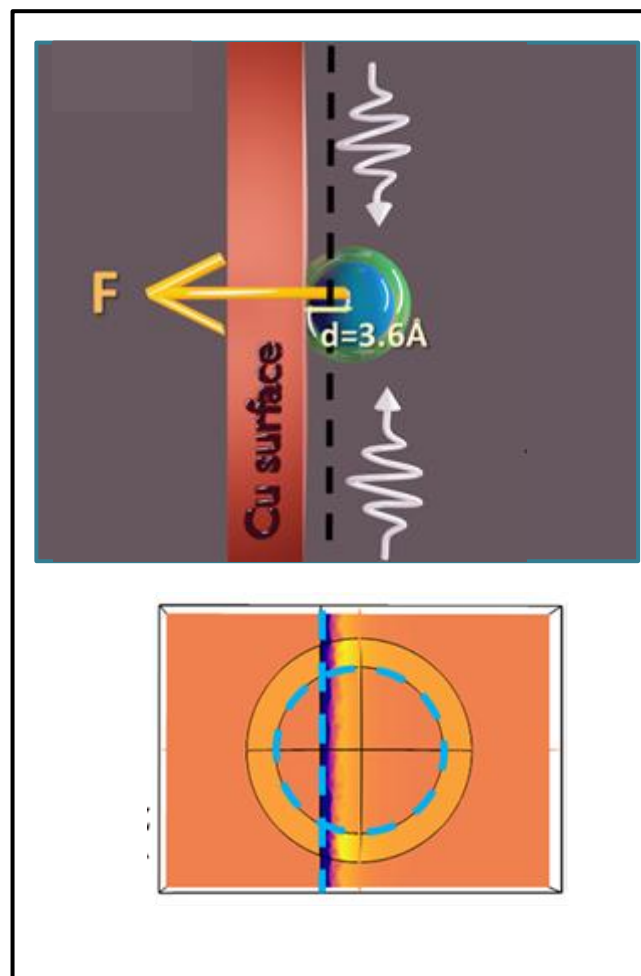
$$\psi_{\text{scattered}} = \frac{f\left(\theta = \frac{\pi}{2}\right)}{|x|} A e^{-ikx}$$

Reflected wave from the metal:

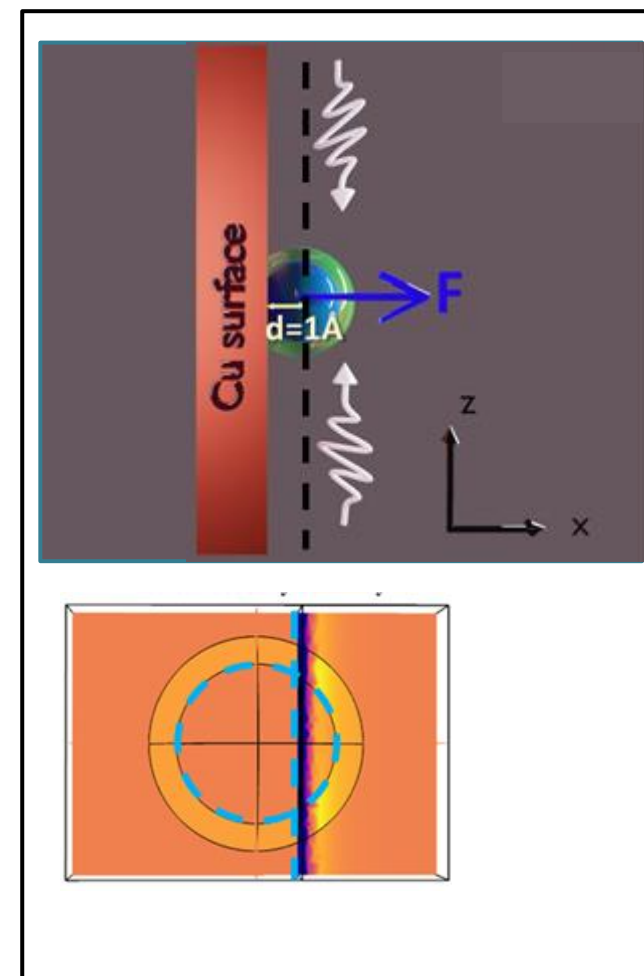
$$\psi_{\text{reflected}} = B e^{ikx} = R_c \frac{f\left(\theta = \frac{\pi}{2}\right)}{|2d - x - \epsilon|} e^{i\delta} A e^{ikx}$$

The total potential energy function is not spherically symmetric, making the calculation non feasible.

Only for $E=10$ meV



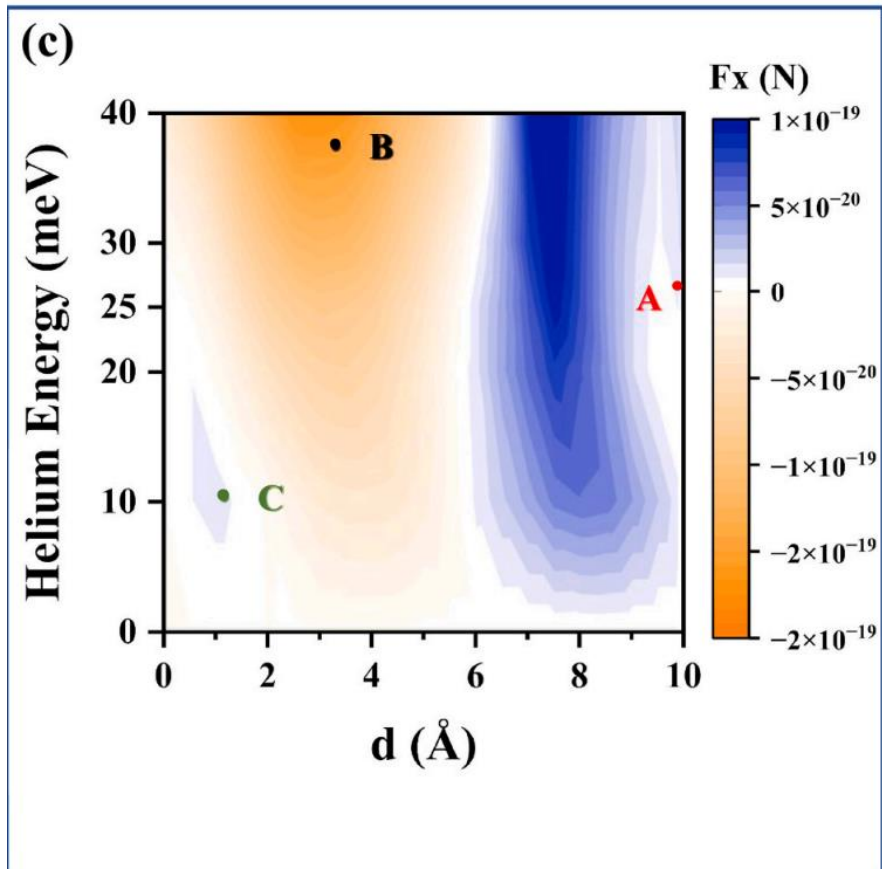
$d = 3.6 \text{ \AA}$



$d = 1 \text{ \AA}$

Adsorption distance for Xenon

Trend in Force Direction



Application: Cleaning the surface

If the target particle gains enough momentum,
it will overcome binding energy and will be desorbed.

Conclusion

To conclude, in this research we have:

- Implemented matter wave scattering in COMSOL Multiphysics and computed force on the scatterer using QM Stress Tensor
- Demonstrated lateral force on a Xenon atom with the help of a metal surface
- Discussed the use of this technique for a cleaning purpose

References

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 2. Salsabil, S. H., Al-Quaderi, G. D., & Mahdy, M. R. C. (2024). Quantum mechanical lateral force on an atom due to matter wave. *Heliyon*, 10(1).
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 4. Mahdy, M.R.C., Al Sayem, A., Shahriar, A. *et al.* Electromagnetic metamaterial-inspired band gap and perfect transmission in semiconductor and graphene-based electronic and photonic structures. *Eur. Phys. J. Plus* 131, 92 (2016).
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Acknowledgement

Acknowledging the support of NSU CTRGC grant 2021–22 and 2022–23 of North South University.

Special thanks to my mentors: Professor Mahdy Rahman Chowdhury and Professor Golam Dastegir Al-Quaderi

Thank you